Homework 5

Collaboration with other students are encouraged; remember to name anyone you collaborate with and to write up your final solution by yourself.

When writing proofs, mathematical clarity is just as important as correctness and rigor. Please keep this in mind and try to adhere to proof writing conventions when typing up solutions.

1. Let
$$G = (V, E)$$
be a connected undirected weighted graph with positive edge weights. Two vertices s and t are given.

   For a path P, define the bottleneck value as
$$\max_{e \in P} w(e),$$
the weight of the heaviest edge on the path.

   Compute a path from s to t minimizing this value
$$\min_{P \in \mathcal{P}(s,t)} \max_{e \in P} w(e),$$
where $\mathcal{P}(s, t)$ is the set of all s-t paths.

   (a) Explain why the following approach does not work and provide a counterexample graph:

   Run Dijkstra's algorithm where the distance of a path is defined as the sum of edge weights.

   (b) Let T be a Minimum Spanning Tree of G. Prove the following statement:

   The path between s and t in the MST minimizes the bottleneck value among all s-t paths in G.

   (c) Using the result from above, design an algorithm that computes the minimum bottleneck path and justify its correctness.

2. We have a directed graph $G = (V, E)$ with positive edge weights and two nodes $u, v \in V$. Suppose we want to find the shortest path from u to v that *passes through* a specified vertex $w \in V$. Design an algorithm that runs in $\mathcal{O}((|V| + |E|) \log |V|)$. Ensure to justify correctness and time complexity. **You should return both the shortest path and its total weight.**

3. Let $G = (V, E)$ be an undirected graph. We say that an edge $e \in E$ is a *cut edge* if removing e from G increases the number of connected components of G. Design a linear time $\mathcal{O}(|V| + |E|)$ algorithm that finds a cut edge of G. Ensure to justify correctness and time complexity. **You should return the cut edge in G, if it exists. Otherwise return nothing.**

4. Suppose you are given a directed graph $G = (V, E)$. Each vertex $v \in V$ is assigned a binary weight $w(v) \in \{0, 1\}$. You are also given a non-negative integer $K \in \mathbb{Z}_{\geq 0}$, representing a total resource budget.

   (a) Design an algorithm to find the length of the shortest path $P = (v_0, v_1, \ldots, v_n)$ from a source vertex s to a target vertex t such that the path satisfies the following resource constraint:
$$\sum_{i=1}^{n} w(v_i) \leq K$$
   Return the minimum number of vertices for which a valid path exists. If no such path exists under the constraint K, return -1.

(b) Prove that if a valid path exists, then your algorithm returns the minimum number of vertices.

(c) Analyze the runtime of your algorithm.

Hint: Reaching a vertex with different amount of resource remaining leads to different future possibilities, so it may be useful to encode resource usage into your visited sets